Nucleotides at Two Timescales: Gated Delta-Rule Linear Attention with k-mer Fast-Weight Memory for DNA Sequence Modeling

A matched-budget ablation of KDA hybrids on the human genome, on a single 8 GB consumer GPU

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ABSTRACT

Gated delta-rule linear attention (Kimi Delta Attention, KDA) has been validated on natural language but, to our knowledge, never on genomic sequence modeling, where the dominant long-context architectures are implicit convolutions (HyenaDNA) and selective state-space models (Caduceus). We pretrain a family of ~2.3M-parameter character-level KDA models on the human reference genome (GRCh38 [10]) at a 131,072 bp training context on a single RTX 5060 (8 GB), and ablate two additions one axis at a time: a per-k-mer *fast-weight memory* that attacks the finite-state recall bound of delta-rule recurrences (D06), and a single windowed local-attention layer plus a sequence-length curriculum (D07). Pure KDA shows no measurable long-range exact recall (copy advantage ≈ 0 beyond 64 bp); the k-mer fast-weight memory recovers strong recall out to 64 kbp (Δ-CE 0.62 nats at distance 65,536). Both arms hold language-model loss essentially flat from 8 K to 1 M bp of context and run 1 Mbp inference in 5.6 GB, whereas released HyenaDNA-1M and Caduceus checkpoints, evaluated through the same harness, OOM at 64 kbp on the same card. On fine-tuned downstream tasks the local window + curriculum arm transforms position-sensitive short-sequence tasks — Nucleotide Transformer splice-site accuracy rises from 0.618 to 0.922 — while *degrading* epigenetic-mark classification (GUE mean −5.3 points), evidence that a fixed ultra-long pretraining length tunes the recurrence's timescales away from sub-kilobase grammar and that the fix is a trade-off, not a free lunch. All comparisons are matched-tokenizer, matched-data, and labelled where pretraining budgets differ; we report negative and mixed results as found.

1 Introduction

DNA foundation models face a context problem that language models do not: regulatory interactions span hundreds of kilobases, yet single-nucleotide resolution matters, so a sequence of interest is routinely 10^5–10^6 tokens long. The two families that dominate this regime — HyenaDNA's implicit long convolutions [3] and the Mamba-based Caduceus [4] — both replace quadratic attention with an O(N·log N) or O(N) mixer. A third

family, *gated delta-rule linear attention* (Gated DeltaNet [2], Kimi Delta Attention [1]), has strong natural-language results but no published genomic evaluation. Its defining weakness is also its most interesting property for genomics: the recurrent state is a fixed d×d matrix, so exact recall of a distant motif is information-theoretically bounded, and prior work formalizes this recall–capacity trade-off [9]. Genomes, full of near-exact repeats whose *differences* are the signal, should be a stress test for exactly this bound.

This paper asks three questions on strictly consumer hardware (one RTX 5060 laptop GPU, 8 GB VRAM): (i) is KDA a viable base operator for a character-level DNA model at 131 kbp training context; (ii) where does its finite-state recall bound show up on genomic tasks, and can it be repaired within an O(N) budget; (iii) what does an ultra-long fixed pretraining context do to sub-kilobase downstream tasks. Contributions:

- **D06 — k-mer fast-weight memory.** A KDA backbone in which every layer also owns a content-addressed memory: each 8-mer address stores a 16×16 fast-weight matrix updated by a gated delta rule. This converts the recall probe from a flat zero (pure KDA) to 0.62–0.87 nats of copy advantage at distances of 64 bp–64 kbp, while remaining O(N) and exact under chunked low-memory inference.
- **D07 — a short-grammar patch.** D06 plus one windowed local-attention layer (w=256) and a 1 K → 131 K sequence-length curriculum. It raises fine-tuned accuracy on 13 of 14 Genomic Benchmarks / Nucleotide Transformer tasks (splice sites +30.4 points) but costs a mean 5.3 points on the 31-task GUE suite — a documented trade-off between short-range grammar and the epigenetic-mark tasks the fixed-length parent was better at.
- **A same-harness efficiency comparison.** Released HyenaDNA and Caduceus checkpoints are run through our identical evaluation harness (same hg38 windows, seeds, probes, bootstrap). Our arms sustain 1 Mbp single-sequence inference in 5.6 GB; the baselines OOM at 64 kbp (Caduceus-PS at 32 kbp) on the same card. Pretraining budgets differ and every such table says so.

Everything here is a matched-compute small-scale study in the spirit of mechanistic ablations, not a claim to beat HyenaDNA at its published scale. Models are ~2.3M parameters; checkpoints analysed are at ≈90% of the 5B-token schedule (§5).

2 Background

Kimi Delta Attention. KDA [1] is a linear-attention recurrence with a channel-wise forget gate, per-head write strength β, ℓ2-normalized queries/keys, and a short causal convolution on q/k/v. The layer maintains a $d_k \times d_v$ state S_t updated by a gated delta rule — an outer-product write that first subtracts what the key currently retrieves, then writes the new value — giving it error-correcting (fast-weight-programmer [8, 9]) semantics at O(N) time and O(1) state per head.

The recall bound. Because S_t has fixed size, the number of (key, value) associations recoverable exactly is bounded; past that, writes interfere. In language this shows up on synthetic multi-query associative recall (MQAR) [9]; in DNA we operationalize it as a repeat-copy probe on real hg38 background (§6.2).

Baselines. HyenaDNA [3] pretrains character-level implicit-convolution models up to 1 M context on hg38; Caduceus [4] is a bidirectional, reverse-complement-equivariant Mamba MLM at 131 K context. We compare against their *released* checkpoints through our harness (§5), never re-implementations.

3 Hypotheses

Following pre-registration practice, the following hypotheses were stated in the study plan before the D06/D07 comparison was run, so that outcomes cannot be reframed after the fact. Each is written as a falsifiable prediction; §6 reports the test for each, and Table 4 (§7) gives the verdicts.

- **H1 (efficiency; high prior confidence).** Linear-complexity KDA arms will process strictly longer sequences than the convolutional and SSM baselines under a fixed 8 GB VRAM ceiling, with peak memory growing linearly in sequence length. *Falsifier:* any baseline matching or exceeding the KDA arms' maximum feasible context on the same card. *Test:* §6.3.
- **H2 (recall bound).** Pure KDA's exact-recall accuracy will degrade past a distance threshold set by its finite d×d state capacity; the k-mer fast-weight memory, which factors exact 8-mer recall out of that state, will hold up at longer range; a windowed local layer will add exact recall only inside its ~256–511 bp span. *Falsifier:* pure KDA matching the memory-augmented arms at long range, or the memory arms degrading identically to pure KDA. *Test:* §6.2.
- **H3 (language-model loss; no prior stated).** Pretraining perplexity across arms was pre-registered as genuinely uncertain, with no directional expectation. *Test:* §6.1.
- **H4 (downstream capability; no prior stated, dissociation expected).** Downstream fine-tuned accuracy was pre-registered as uncertain, with the explicit caution that perplexity may not predict it. The D07-specific sub-hypothesis, formed after D06's splice-site failure and before D07 was trained: a local-attention window plus a length curriculum will repair sub-kilobase position-sensitive tasks that fixed-131 K pretraining damaged. *Falsifier:* D07 failing to close the gap to the random-init control on splice sites. *Test:* §6.5.

4 Architecture

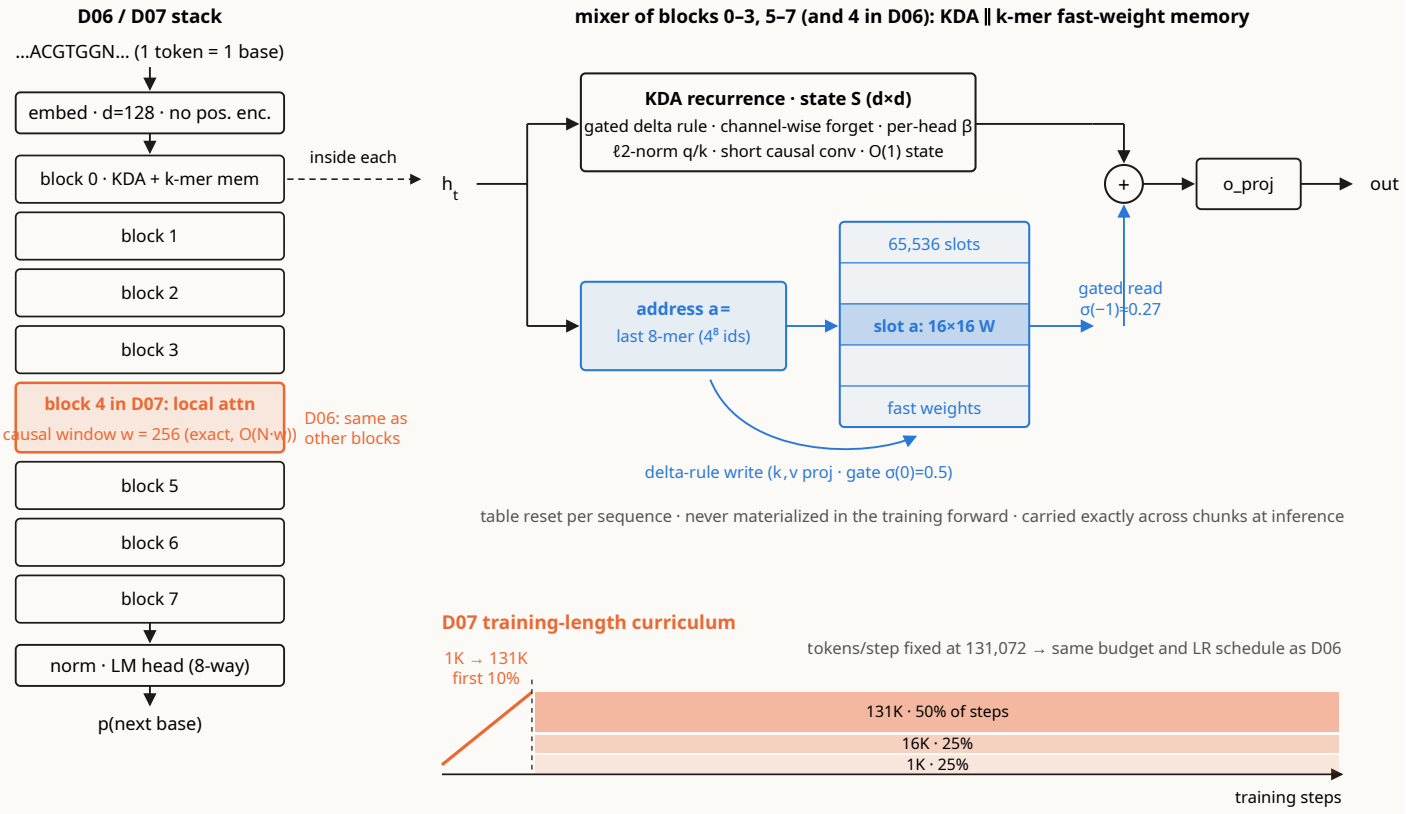


Figure 1 — The D06/D07 stack and the per-layer k-mer fast-weight memory. Left: eight pre-norm transformer blocks, d_{model}=128, no positional encoding; every block's mixer is KDA plus the k-mer memory. In D07 only, block 4's mixer is replaced by blocked causal local attention (window 256, effective span 256–511 bp). Right: the memory path inside each mixer — the last 8 nucleotides address a table slot holding a 16×16 fast-weight matrix; a gated delta rule writes through W_k/W_v projections, and a gated read is added to the KDA output before o_proj. The table is reset per sequence and carried exactly across chunks at inference.

4.1 D06: KDA + per-k-mer fast-weight memory

The backbone is 8 layers, d_{model}=128, 4 heads, vocabulary of 8 (A/C/G/T/N + specials), no positional encoding (NoPE), 2.32M parameters. Alongside each layer's KDA mixer sits a content-addressed memory keyed by the exact identity of the last k=8 nucleotides — 4^8=65,536 possible addresses. Each address stores not a vector but a small delta-rule fast-weight matrix (d_k=d_v=16): when position t's 8-mer address is a, the layer writes (k_t, v_t) projections of the hidden state into slot a's matrix with a learned write gate (bias init 0.0, i.e. σ=0.5), and reads slot a's matrix with the current key under a learned read gate (bias init −1.0, σ≈0.27), adding the result to the mixer output pre-`o_proj`. Two properties matter:

- **It targets the recall bound directly.** Recall of "what followed this exact 8-mer last time" no longer competes for the KDA state's fixed capacity: the address space factors it out, and the delta rule inside each

slot handles collisions of repeated 8-mers.

- **It stays O(N) and chunk-exact.** The full table (≈0.6 GB per training sequence) is never materialized during the training forward — only chained per-chunk carries are computed — and chunked inference carries the table across chunks bitwise-exactly, which is what makes the 1 Mbp low-memory evaluation path legitimate (§6.3).

Two training details were load-bearing rather than incidental: gate parameters are kept in fp32 (scalar gates freeze in bf16), and the read/write gate initializations above came from a bias sweep after an early bug zeroed the write gate.

4.2 D07: one local-attention layer + a sequence-length curriculum

D07 is D06 with two additions, each motivated by a specific observed failure (§7): a *random-init* D06 fine-tuned with a locus-aware head reached 0.86 on NT splice-sites while the 4.5B-token 131 K-context checkpoint reached 0.82 — pretraining at a fixed 131 K length had tuned the recurrence's timescales *away* from 300–600 bp grammar.

- **Local window.** Block 4's mixer becomes blocked causal local attention with window 256 (effective causal span 256–511 bp): exact token-level recall inside the window regardless of learned timescales. Window 256 rather than 512 was chosen so that a 300–600 bp task sequence is not resolvable in a single exact hop.
- **Length curriculum.** Training length warms up 1 K → 131 K over the first 10% of steps, then draws per-step lengths from a fixed mix {1 K: 0.25, 16 K: 0.25, 131 K: 0.5}. Tokens per step are unchanged (131,072), so the token budget and LR schedule are identical to D06. A loss-driven length controller was considered and rejected: bang-bang switching oscillates and is not reproducible across arms; the fixed mix keeps both regimes in the loss by construction, monitored by `val_loss` (131 K) and `val_loss_short` (1 K).

Micro-batching is capped by tokens in flight, not sequences (4×32,768 peaks at 6.17 GiB where 1×131,072 peaks at 4.35 GiB), which is what lets the curriculum train on the 8 GB card. D07 is 2.29M parameters.

Arm	Mixer	Params	Pos.	Train ctx	Extras
D04	pure KDA	2.25M	NTK-RoPE	65,536	—
D06	KDA + k-mer fast-weight memory	2.32M	none	131,072	—
D07	D06, block 4 → local attn (w=256)	2.29M	none	131,072	length curriculum

Table 1 — the arms analysed in this paper. Earlier ablation arms (D01–D05: pure KDA, windowed hybrids, KATA, evicted-KV cache, vector-valued k-mer memory) established the design and are referenced where relevant; D04 serves here as the pure-KDA recall control.

5 Experimental setup

Pretraining. Human reference genome GRCh38/hg38 [10], single-nucleotide tokenization (vocab 8), HyenaDNA's held-out-chromosome split, causal next-nucleotide objective. Budget 5B tokens (38,147 steps × 131,072 tokens), AdamW, lr 10^{-3} cosine to 0.1×, warmup 200 steps, bf16 with gradient checkpointing, batch 1×131,072 tokens/step. Hardware: one NVIDIA RTX 5060 Laptop GPU (8 GB, Blackwell). **The checkpoints analysed are at step 34,177 (D06) and 33,900 (D07) — 89–90% of the schedule**; training of the final 10% was still running when this draft was written, and all numbers below are from these ≈90% checkpoints.

Evaluation harness. Every number comes from one suite (`scripts/benchmark.py`): per-(benchmark, context) reseeding of numpy+torch, a 4M-token floor for LM evaluation per context, 128 trials per needle point, and a percentile bootstrap (n=1000) on every test-set metric, resampling per prediction (classification), per window (LM), or per trial (needle) — split-sampling uncertainty only, not seed variance. Downstream tasks run in two modes: a frozen-backbone linear probe (lower bound) and full fine-tuning (protocol family of HyenaDNA/NT/DNABERT-2; effective batch 32, up to 6 epochs / 6,000 steps, best-dev-epoch selection, a `[mean | last | 8 positional bins]` head pooled over forward and reverse-complement passes). External baselines — `hyenadna-large-1m`, `hyenadna-medium-450k`, `caduceus-ph/ps 131k` — run through the same harness with vocabulary remapped losslessly; their pretraining budgets are far larger than ours, and every table containing them is labelled accordingly. Caduceus is a bidirectional MLM, so causal LM/needle metrics are skipped for it by design. The chunked low-memory path (`--offload`) is used only for our KDA arms, where it is verified bitwise-exact; the baselines' OOM ceilings are real measurements, not artifacts of a missing offload path.

6 Results

6.1 Pretraining (tests H3): matched loss, flat across three orders of magnitude of context

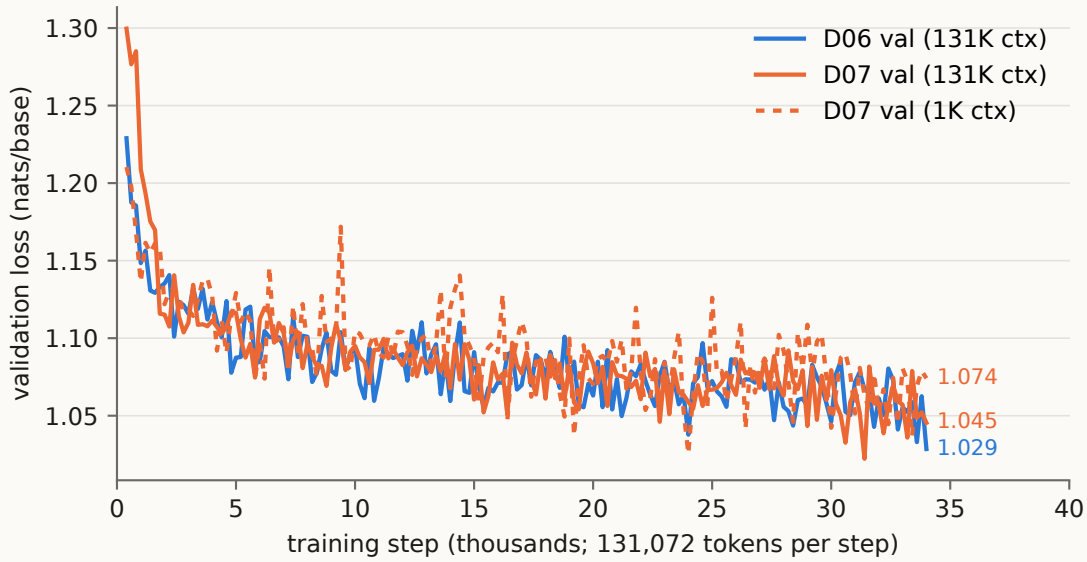


Figure 2 — Validation loss. D06 and D07 track each other closely at the shared 131 K evaluation length (final 1.029 vs 1.045 nats/base); D07 additionally holds a 1 K-context validation loss within 0.03 nats of its long-context loss, the curriculum's explicit target. Curves start after LR warmup.

At the ≈90% checkpoints, held-out cross-entropy is 1.065 (D06) and 1.067 (D07) nats/base at 131 K context — 1.54 bits/bp, perplexity 2.90 over the 4-letter alphabet — and stays within ±0.03 nats from 8 K to 1 M bp of evaluation context for both arms. The local window and curriculum are loss-neutral at long range: whatever they change (§6.5), it is not next-nucleotide modeling of megabase windows.

6.2 The recall bound (tests H2) is real, and the k-mer memory removes most of it

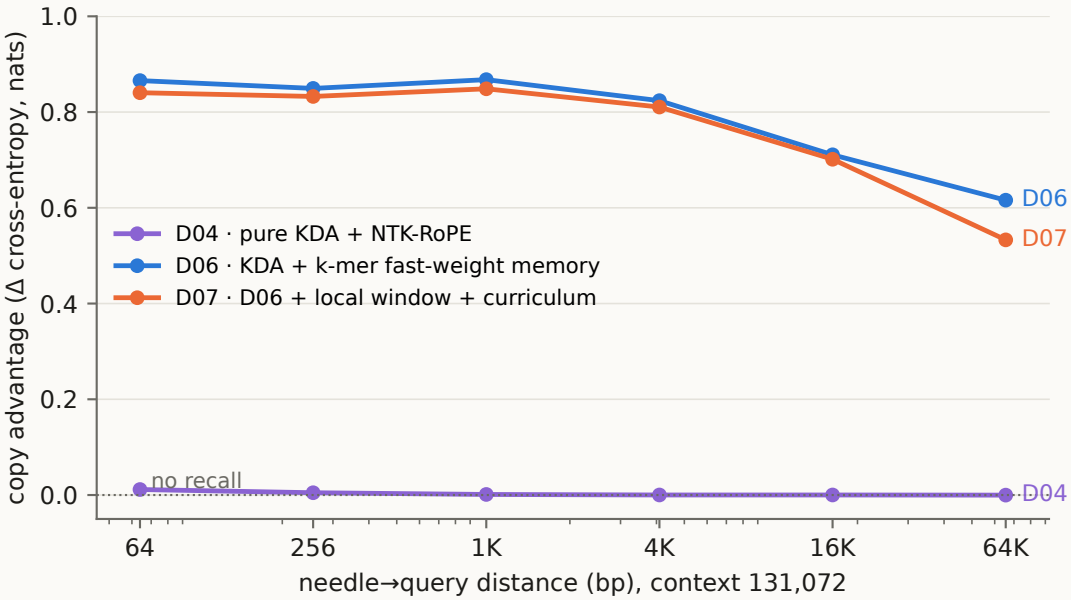


Figure 3 — Long-range exact recall (context 131,072). A 256 bp segment of real hg38 sequence is planted, the model must lower its loss when the segment repeats a distance d later; the y-axis is the paired per-trial CE reduction versus a control (128 trials/point). Pure KDA (D04) is indistinguishable from zero beyond 64 bp — the finite-state bound in its rawest form. The k-mer fast-weight arms hold 0.53–0.87 nats of advantage out to 64 kbp, decaying smoothly with distance.

This is the mechanistic centerpiece for H2 and it resolves cleanly. H2 predicted that pure KDA would degrade past a capacity threshold: in practice there is no threshold to find — D04's copy advantage is ≈0.01 nats at distance 64 and zero beyond, i.e. pure KDA never performs measurable exact recall at any distance on genomic background. The fast-weight memory (D06) recovers 0.87 nats at short range, still 0.62 at 64 kbp. D07 sits slightly below D06 at long distances (0.53 at 64 kbp) — its local window contributes exact recall only inside ~500 bp, and the curriculum's short-sequence steps plausibly de-emphasize ultra-long carry — but the two arms are qualitatively identical. An interesting negative within the positive: the decay from 0.85 to 0.53 nats between 4 K and 64 K is interference inside slot-level fast weights, not address exhaustion, since address collisions are distance-independent.

6.3 Efficiency (tests H1): megabase inference on 8 GB

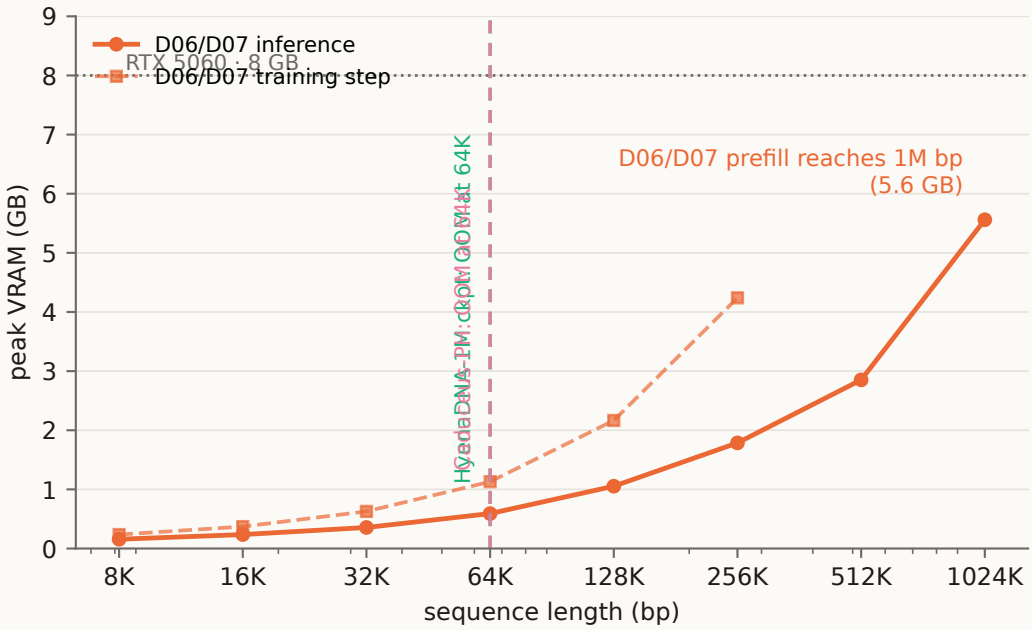


Figure 4 — Peak VRAM versus sequence length (batch 1, bf16). D06/D07 inference is linear and reaches 1 Mbp at 5.6 GB; a full training step fits through 256 K. Vertical dashed lines mark where the released HyenaDNA-1M and Caduceus-PH checkpoints OOM in the same harness on the same card (their pretraining used far larger hardware; the ceiling shown is what an 8 GB consumer deployment of those checkpoints gets). Prefill throughput for D06/D07 is 290–350 K tok/s across the sweep.

H1 was the high-confidence pre-registration and it holds with a caveat worth stating: the k-mer table makes our arms *less* memory-light than pure KDA (D04 runs a 1 Mbp forward in 6.3 GB without offload; D06/D07 need the chunk-exact offload path above 512 K to keep the needle sweep alive). Still, on the metric that matters for consumer deployment — the longest single sequence processable at all — our arms reach 1 Mbp where HyenaDNA-1M reaches 32 K (OOM at 64 K) and Caduceus-PS 16 K (OOM at 32 K).

6.4 Frozen-probe biology: long inputs help until they don't

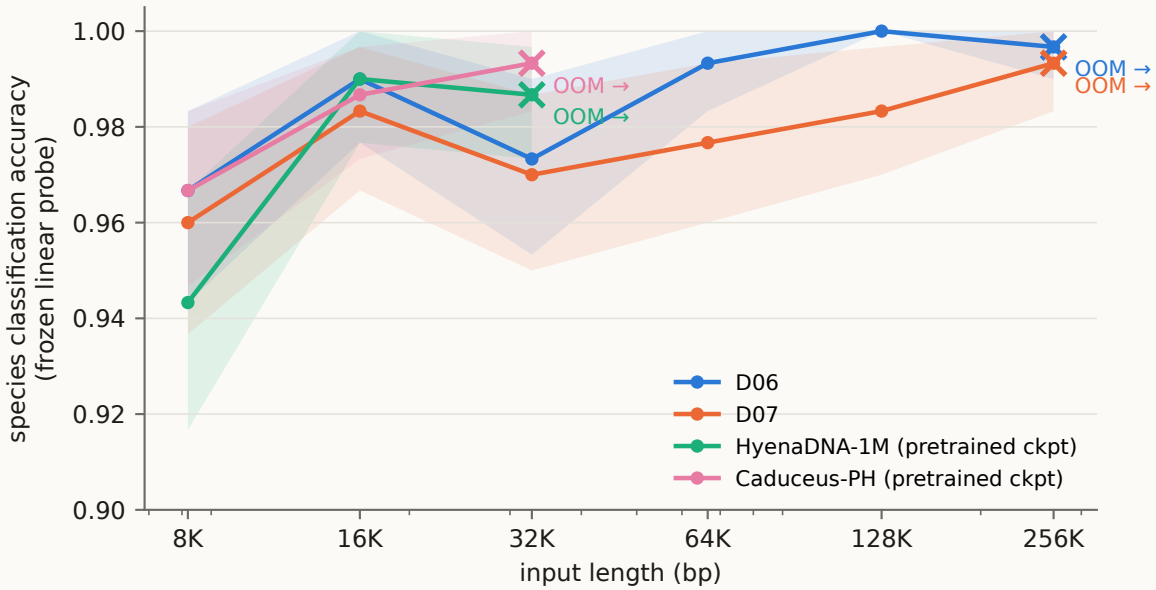


Figure 5 — Species classification versus input length (frozen linear probe, 5-way: human/mouse/zebrafish/fly/worm; shaded bands are 95% bootstrap CIs; × marks the last length before OOM). All models saturate near 0.99 by 16 K; the differentiator is how far each can go — D06 reaches 256 K windows at 0.997, the external checkpoints stop at 16–32 K. External baselines have much larger pretraining budgets.

Task (probe, 8 K windows)	D06	D07	D02	HyenaDNA-1M	Caduceus-PH	Caduceus-PS
GLRB VEP eQTL (acc)	**0.672**	0.625	0.664	0.586	0.586	**0.672**
GLRB enhancer (acc)	0.608	0.648	**0.664**	0.592	0.624	0.632
GLRB promoter (acc)	0.942	0.942	**0.950**	0.934	0.934	0.934
GLRB chromatin (mean acc)	0.931	0.937	0.929	0.933	0.937	**0.938**

Table 2 — Genomics Long-Range Benchmark, frozen-backbone linear probes at 8 K context, same harness for every column. A probe is a lower bound on all models equally. External checkpoints (right three columns) were pretrained with orders-of-magnitude more compute than ours; typical 95% CIs span ±0.07 on eQTL/enhancer, so most differences here are individually indistinguishable — the honest summary is parity, achieved at ~2.3M parameters and one consumer GPU of pretraining.

Longer inputs help on species identity (monotone to 256 K) but *hurt* on GLRB variant-effect prediction, where every model degrades from 8 K to 32 K windows (D06: 0.672 → 0.578) — diluting a SNP-centered signal with context is not free. "More context" is a per-task question, and the harness answers it per task rather than assuming the long-context story is uniformly positive.

6.5 Fine-tuned downstream (tests H4): the trade D07 makes

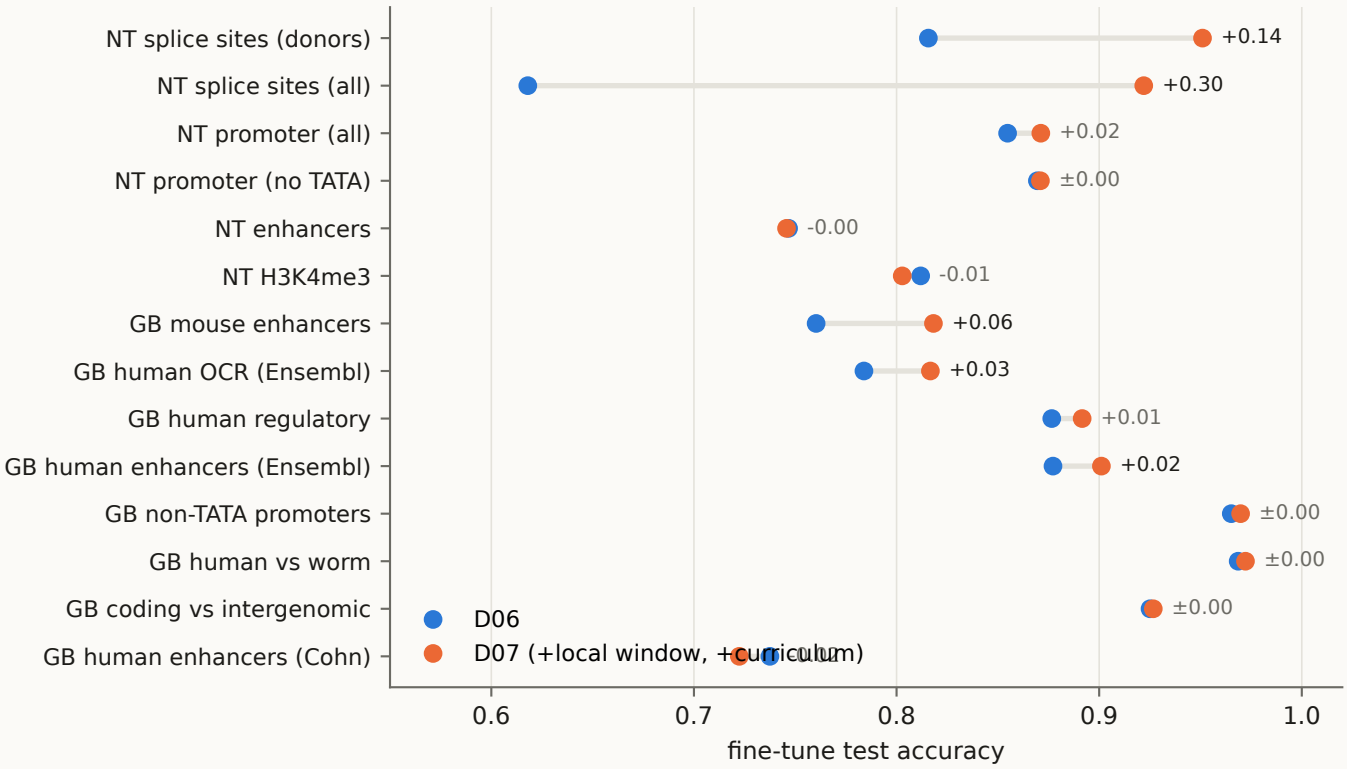


Figure 6 — Full fine-tuning, D06 → D07, Genomic Benchmarks (8 tasks) and the 6-task Nucleotide Transformer subset. D07 improves 13 of 14 tasks; the splice-site tasks move +30.4 and +13.5 points (n_{test}=3,000 each, so the gaps are far outside bootstrap noise).

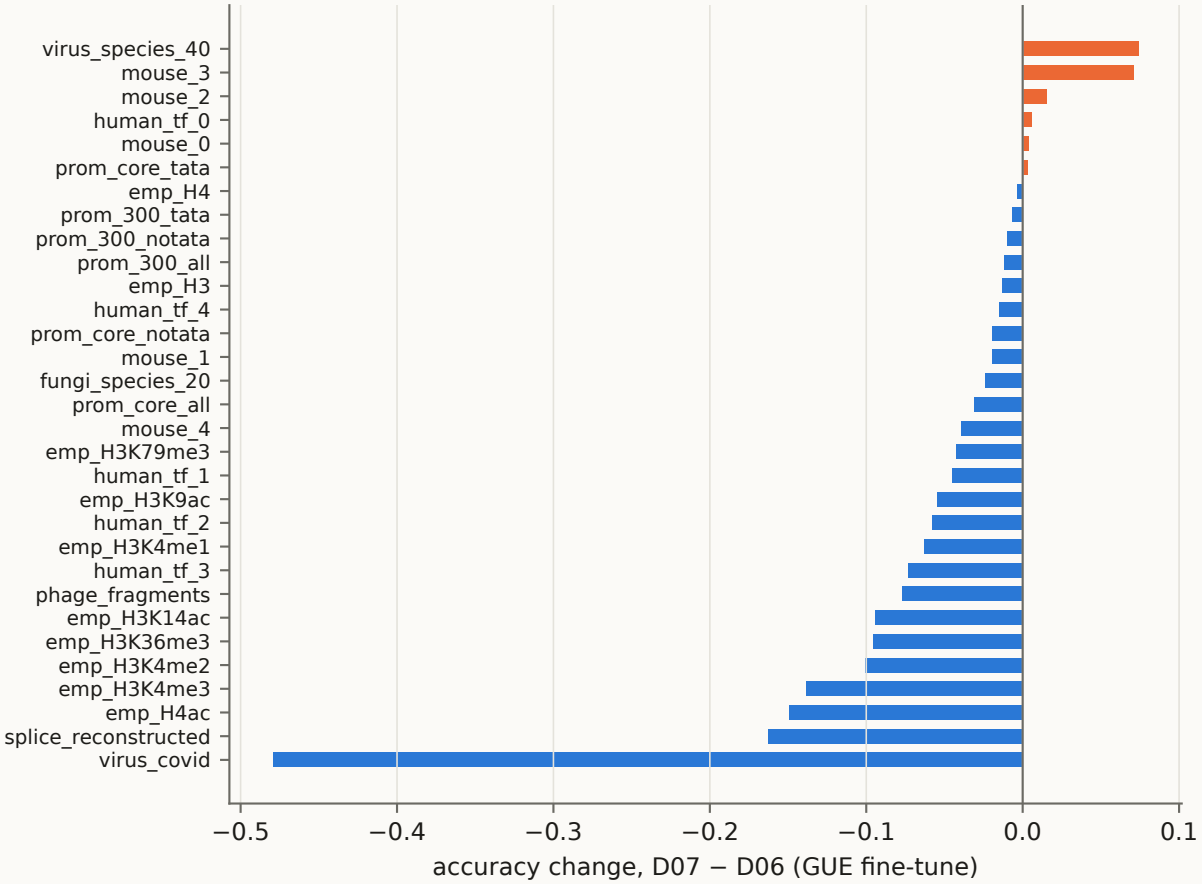


Figure 7 — The other side of the ledger: per-task accuracy change on the 31-task GUE suite. Orange bars (D07 better) are concentrated in mouse TF-binding and promoter tasks; blue (D06 better) dominates the ten `emp_*` yeast epigenetic-mark tasks (−4 to −14 points) and `virus_covid` collapses outright (0.657 → 0.177, a 9-class task where D07 fine-tuning failed to converge under the fixed budget).

Suite (fine-tune, mean acc)	#tasks	D06	D07	Δ
Genomic Benchmarks	8	0.862	**0.877**	+1.6
Nucleotide Transformer subset	6	0.786	**0.861**	+7.5
GUE (DNABERT-2)	31	**0.802**	0.749	−5.3

Table 3 — suite-level means. GUE data is the `leannmlindsey/GUE` HuggingFace community mirror of the DNABERT-2 release (the official Google-Drive distribution cannot be fetched non-interactively).

The headline result of the project sits here. D06's worst downstream failure — splice sites, where its 0.618 fine-tuned accuracy fell *below a randomly-initialized copy of itself* (0.86) — was direct evidence that 131 K-only pretraining damages sub-kilobase positional grammar rather than merely not helping it. D07's window-plus-curriculum repairs exactly that class of task: splice donors 0.951, splice all 0.922, now well above the random-init control, plus consistent gains on enhancer/OCR/regulatory-element classification. But the repair is paid for on GUE's epigenetic-mark tasks, which are dominated by soft compositional signals over 500 bp; one plausible reading is that block 4's exact local recall displaces mixer capacity these tasks were using, and the curriculum's

short-sequence gradient steps re-tune timescales that the `emp_*` tasks preferred long. Perplexity predicted none of this — the two arms are 0.002 nats apart (§6.1) and 5–30 points apart per task in both directions, this domain's restatement of "perplexity is not capability."

7 Discussion and limitations

Hypothesis (§3)	Pre-registered prediction	Verdict	Evidence
H1 efficiency	KDA arms exceed baselines' max context on 8 GB	**Supported**	1 Mbp vs 32 K/16 K before OOM (Fig. 4, Fig. 5)
H2 recall bound	Pure KDA degrades past a distance threshold; memory arms hold longer	Supported, stronger than predicted	Pure KDA has *no* recall at any distance; memory arms hold to 64 kbp (Fig. 3)
H3 LM loss	No direction stated	Parity	D06/D07 within 0.002 nats; flat 8 K–1 M (Fig. 2)
H4 downstream	Perplexity–capability dissociation; window+curriculum repairs sub-kb tasks	Supported, with an unpredicted cost	Splice +30.4 pts, above random-init control; GUE −5.3 mean (Figs. 6–7)

Table 4 — verdicts on the pre-registered hypotheses. "Supported" means the falsifier stated in §3 did not occur; H4's GUE regression was not predicted in either direction and is reported as a discovered trade-off, not a confirmation.

- **The composite is not yet decomposed.** D07 changes two things over D06. The single-axis controls exist as configs (`d06_curriculum.yaml`: curriculum without window; D06 itself: neither) but the curriculum-only control was untrained at this draft's checkpoint; attributing the splice-site gain between window and curriculum — and the GUE loss likewise — is the single most important pending run.
- **≈90% checkpoints, one seed.** Numbers are from step ~34 K of 38,147, single seed, single data order. Loss curves are flat enough that the last 10% is unlikely to reorder arms, but final-checkpoint numbers will replace these; bootstrap CIs cover split-sampling uncertainty only.
- **Capacity ceiling.** At d_{model}=128, NT `promoter_all` is pinned at ~0.85–0.87 under every head, learning rate, and initialization tried (GC content alone yields 0.81) — a model-capacity ceiling, reported as such. The ~2.3M-parameter scale (chosen for training throughput on the 8 GB card, well below the 100M originally envisioned) bounds every accuracy claim.
- **Comparisons with released baselines are efficiency comparisons.** HyenaDNA/Caduceus columns share our harness but not our pretraining budget; where they match or beat us the reading is unremarkable, where we match them at 2.3M parameters the reading is efficiency, and their OOM ceilings describe 8 GB deployment of those checkpoints, not the methods at their native scale.

- **What we would test next.** The curriculum-only control; finishing the last 10% and regenerating all tables; a g/window-size sweep of the local layer; and state-utilization diagnostics (participation ratio of the recurrent state) to localize the 4 K→64 K recall decay inside slot fast weights.

8 Conclusion

Gated delta-rule linear attention is a workable base operator for character-level genomic modeling on consumer hardware — but not alone. Pure KDA's finite-state recall bound is not a distant cliff on DNA; it is a wall at distance zero. A content-addressed k-mer fast-weight memory removes most of that wall for ~3% extra parameters while preserving O(N) scaling and megabase-length exact chunked inference on an 8 GB GPU, which the released convolutional and SSM checkpoints cannot match on the same card. And the fixed ultra-long pretraining context that makes such models impressive at length makes them *worse* at sub-kilobase grammar than their own random initialization — a failure a single local-attention layer and a length curriculum repair, at a measurable cost elsewhere. The trade-offs, not the wins, are the finding: recall, context length, and short-range grammar are three different budgets, and an architecture choice that spends one is spending it from somewhere.

9 References

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